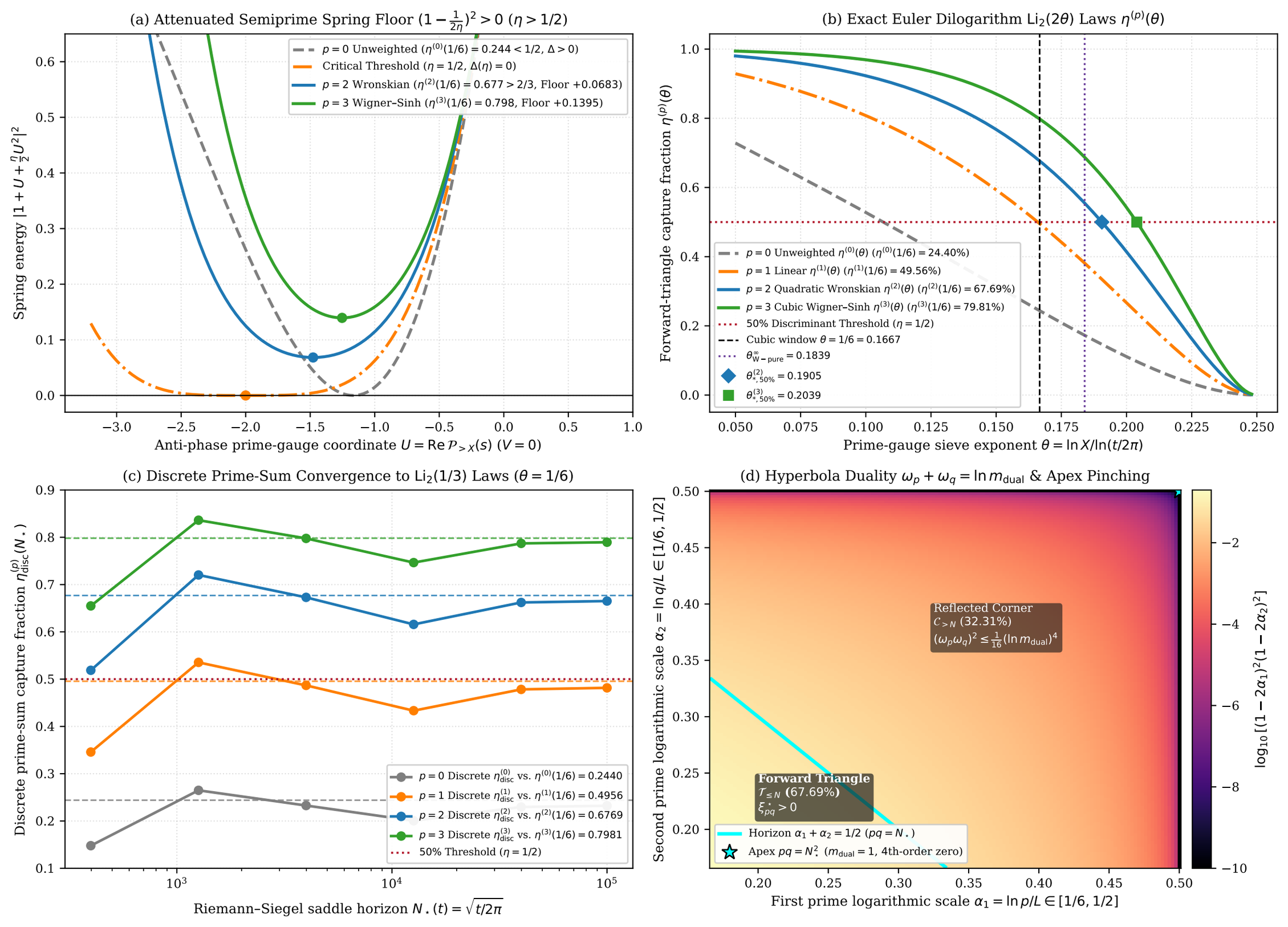

(a) Attenuated Semiprime Spring Floor $(1-\frac{1}{2\eta})^2 > 0$ $(\eta > 1/2)$
$p = 0$ Unweighted ($\eta^{(0)}(1/6) = 0.244 < 1/2$, $\Delta > 0$)
Critical Threshold ($\eta = 1/2$, $\Delta(\eta) = 0$)
$p = 2$ Wronskian ($\eta^{(2)}(1/6) = 0.677 > 2/3$, Floor +0.0683)
$p = 3$ Wigner–Sinh ($\eta^{(3)}(1/6) = 0.798$, Floor +0.1395)
Spring energy $|1 + U + \frac{\eta}{2}U^2|^2$
Anti-phase prime-gauge coordinate $U = \mathrm{Re}\,\mathcal{P}_{>X}(s)$ $(V = 0)$
(b) Exact Euler Dilogarithm $\mathrm{Li}_2(2\theta)$ Laws $\eta^{(p)}(\theta)$
$p = 0$ Unweighted $\eta^{(0)}(\theta)$ ($\eta^{(0)}(1/6) = 24.40\%$)
$p = 1$ Linear $\eta^{(1)}(\theta)$ ($\eta^{(1)}(1/6) = 49.56\%$)
$p = 2$ Quadratic Wronskian $\eta^{(2)}(\theta)$ ($\eta^{(2)}(1/6) = 67.69\%$)
$p = 3$ Cubic Wigner–Sinh $\eta^{(3)}(\theta)$ ($\eta^{(3)}(1/6) = 79.81\%$)
50% Discriminant Threshold ($\eta = 1/2$)
Cubic window $\theta = 1/6 = 0.1667$
$\theta^{\infty}_{W-\mathrm{pure}} = 0.1839$
$\theta^{(2)}_{*,50\%} = 0.1905$
$\theta^{(3)}_{*,50\%} = 0.2039$
Forward-triangle capture fraction $\eta^{(p)}(\theta)$
Prime-gauge sieve exponent $\theta = \ln X/\ln(t/2\pi)$
(c) Discrete Prime-Sum Convergence to $\mathrm{Li}_2(1/3)$ Laws $(\theta = 1/6)$
$p = 0$ Discrete $\eta^{(0)}_{\mathrm{disc}}$ vs. $\eta^{(0)}(1/6) = 0.2440$
$p = 1$ Discrete $\eta^{(1)}_{\mathrm{disc}}$ vs. $\eta^{(1)}(1/6) = 0.4956$
$p = 2$ Discrete $\eta^{(2)}_{\mathrm{disc}}$ vs. $\eta^{(2)}(1/6) = 0.6769$
$p = 3$ Discrete $\eta^{(3)}_{\mathrm{disc}}$ vs. $\eta^{(3)}(1/6) = 0.7981$
50% Threshold ($\eta = 1/2$)
Discrete prime-sum capture fraction $\eta^{(p)}_{\mathrm{disc}}(N_\star)$
Riemann–Siegel saddle horizon $N_\star(t) = \sqrt{t/2\pi}$
(d) Hyperbola Duality $\omega_p + \omega_q = \ln m_{\mathrm{dual}}$ & Apex Pinching
Reflected Corner
$\mathcal{C}_{>N}$ (32.31%)
$(\omega_p\omega_q)^2 \le \frac{1}{16}(\ln m_{\mathrm{dual}})^4$
Forward Triangle
$\mathcal{T}_{\le N}$ (67.69%)
$\xi^{\star}_{pq} > 0$
Horizon $\alpha_1 + \alpha_2 = 1/2$ $(pq = N_\star)$
Apex $pq = N_\star^2$ ($m_{\mathrm{dual}} = 1$, 4th-order zero)
Second prime logarithmic scale $\alpha_2 = \ln q/L \in [1/6, 1/2]$
First prime logarithmic scale $\alpha_1 = \ln p/L \in [1/6, 1/2]$
$\log_{10}[(1-2\alpha_1)^2(1-2\alpha_2)^2]$